



Fig. 13: Component layout for the PCB shown in Fig. 12

tion software such as Terminal v1.9b, select the com port and make the aforementioned settings. After completing all the settings and component assembly, switch on the power supply to the transmitter and receiver circuits. Run Terminal v1.9b software and activate the connection by pressing 'Connect' button followed by W, S, D, A, Q, E, C, Z or U key on the keyboard. The ASCII code corresponding to that key will be transmitted serially in a 10-bit (1 start bit+8 data bits for ASCII code+1 stop bits) binary datastream. The toy car will move as per the input from the keyboard.

EFY note. The source code of this project has been included in this month's EFY-CD and is also available on www.efymag.com website.

TOYCAR.C

```
#include<reg51.h>
sbit mla = P1^1; //output 1 (Q1)
sbit mlb = P1^2; //output 2 (Q2)
sbit m2a = P1^3; //output 3 (Q3)
sbit m2b = P1^4; //output 4 (Q4)
void RXDATA(unsigned char) unsigned
char RX(void); void MSDELAY(unsigned
char); void A(); void S(); void D(); void
W();
void Z(); void C(); void U(); void E(); void
Q();

void main()
{
    unsigned char a;
    SCON=0x50;
    TMOD=0x20;
    TH1=-24;
    TR1=1;
    PCON=0x40;
    P1=0x00;
    while(1)
    {
        a=RX();
        RXDATA(a);
    }
}

void RXDATA(unsigned char a) //Re-
ceiving a byte data through serial port
{
    switch(a)
    {
        case 'A':
            A();
            break;
        case 'S':
            S();
            break;
        case 'D':
            D();
            break;
        case 'W':
            W();
            break;
        case 'Z':
            Z();
            break;
        case 'Q':
            Q();
            break;
        case 'E':
            E();
            break;
        case 'U':
            U();
            break;
    }
}

void A() //Left drift
{
    unsigned char i;
    P1=0;
    m2a=1;
    for(i=0;i<9;i++)//PWM for 1/3 duty
    cycle
    {
        mla=1;
        MSDELAY(15);
        mla=0;
        MSDELAY(30);
    }
    P1=0;
}

void S() //Backward movement
{
    P1=0;
    mlb=1;
    m2b=1;
}

void D() //Right drifts
{
    unsigned char i;
    P1=0;
    mla=1;
    for(i=0;i<9;i++)//PWM for 1/3 duty
    cycle
    {
        m2a=1;
        MSDELAY(15);
        m2a=0;
        MSDELAY(30);
    }
    P1=0;
}

void Z() //Backward left turn
{
    P1=0;
    mlb=1;
    m2b=1;
}

void C() //Sharp backward right turn
{
    P1=0;
    m2b=1;
}

void Q() //Sharp forward left turn
{
    P1=0;
    m2a=1;
}

void E() //Sharp forward right turn
{
    P1=0;
    mla=1;
}

void U() //Stop
{
    P1=0;
}

unsigned char RX() // Function
to receive a data
{
    EMERGENCY;
    RI=0;
    while(RI==0);
    if(SMO==1)//Check for framing
    error
    {
        SMO=0;
        goto EMERGENCY;
    }
    return(SBUF);
}

void MSDELAY(unsigned char b) //Delay
of 3.6ms
{
    unsigned char i;
    unsigned int j;
    for(i=0;i<=b;i++)
    for(j=0;j<=310;j++);
}
```